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Player-Wage Efficiency in the Premier League: Isolating and Testing the Persistence of Performance Unexplained by Player Wages (2019/20–2025/26)

Abstract:

Spending and success in club football display a well-established link in the long run: higher wage payrolls are associated with better performance. However, averaging over many seasons mutes the season-level performance that player wage bills leave unexplained. Using 140 *Premier League* club-seasons (2019/20–2025/26), this study regresses league points on log player wages with season fixed effects and cluster-robust standard errors, then models the resulting residuals as a first-order autoregression against each club's previous season residual. Wage bills explain ~47% of single-season points variation, with a doubling of wage bills being associated with ~15 additional points. Persistence analysis reveals that ~38% of a club's over- or under-performance carries into the next season ($p < 0.001$), indicating that durable non-wage factors do exist, though the low R^2 (0.147) means most of any single season's deviation from the wage-derived points prediction remains transient. *Brentford*, *Manchester City* and *Liverpool* emerge as the most persistently efficient, on average beating wage-derived predictions by 13-16 points per season; *Southampton*, *Manchester United* and *Everton* the least, underperforming by 8-11 points. As player wages alone were used, the residual absorbs non-player spending, non-wage attributes and possible persistent measurement error: results indicate the existence of a relationship rather than an exact measurement, and the specific durable non-wage factors cannot be isolated.

Introduction:

The notion that money buys success in football is entrenched in the zeitgeist to the point that it seems trivial to state: Hall et al. (2002) concluded that causality between higher payrolls and better performance in English football cannot be rejected, with Szymanski (2021) quantifying that, over the long-run, ~90% of the variation in average league position in England's top two divisions can be explained by average wage spending. Szymanski himself notes that this relationship is far less reliable in the short-term, where luck, injuries and runs of form make single seasons noisier; with good and bad fortune cancelling out over time. It is worth noting that averaging over a decade mutes club-specific operational advantages, which are often short-lived: due to football's dynamic nature, these 'alphas' tend to be competed out of existence or absorbed into other teams' philosophies within a few seasons. The choice of approach depends entirely on the research objective and it is precisely this short-term variation that this study sets out to retain and examine. Retaining the season-level variation costs explanatory power (a lower R^2) but it is what makes it possible to identify which clubs out- and under-perform their season's wage bill, and to test whether that performance persists.

This study aims first to determine the extent to which player wages can predict a single season's points tally, and then to explore the prediction errors: the performance not explained by wage spending. This 'wage efficiency' can then be examined to reveal whether it is merely due to chance or driven by durable club-level factors, such as coaching quality, recruitment strategy and facilities. The most 'wage-efficient' and 'wage-inefficient' clubs can then be identified to give a gauge of the *Premier League* clubs that are the most, and least, operationally astute.

Data:

This study uses league points and annual player-wage data spanning seven seasons of the *Premier League*, from *2019/20* to *2025/26*, with league points sourced from *FBref* (n.d.), a publicly available football statistics database, and wage data from *Capology,* which specialises in professional player salaries: drawing on public information; industry experts; a network of insiders, and algorithmic estimation. 'Verified' salaries are defined as those that are provided directly by the club, agent, intermediary, or confirmed by 2 or more sources; unverified figures are algorithmic estimates (Capology, n.d.). Capology has applied this verification method since the 2019/20 season, meaning all earlier wage bills are 100% estimated (Lynch, 2022), providing the impetus for limiting the study to *2019/20* onwards.

On their website, *Capology* states that wage bill accuracy is correlated with league popularity, owing to information availability and press coverage (Capology, n.d.). This study therefore concentrates on the *Premier League*, which is comfortably the largest and most heavily covered of Europe's 'big-five' leagues: its clubs generated €8.1 billion in revenue in 2024/25, roughly double the next-largest league; with average revenue per club of €405m against €236m in the Bundesliga and €120m in Ligue 1 (Deloitte, 2026, Chart 2). Broadcast revenue alone reached €4.0 billion: more than 2x that of any other 'big five' European league; reflecting a scale of scrutiny that increases the volume of reporting and the number of sources against which salaries can be verified.

Table 1: Descriptive Statistics (n = 140 club-seasons)

Variable	N	Mean	SD	Min	Median	Max
League Points	140	52.42	17.83	12.00	52.00	99.00
Wage Bill (£m)	140	90.48	51.20	20.09	73.24	238.85
ln(Wage Bill)	140	18.16	0.57	16.82	18.11	19.29

The panel covers 28 different clubs across seven *Premier League* seasons, giving 140 club-seasons. The wage distribution is heavily right-skewed: the mean bill (£90.5m) exceeds the median (£73.2m), and the largest is 11.9 times the smallest.

Further considerations: the degree of estimation in player salaries varies by club and by season. Any specific, quantifiable result from this study should be interpreted as indicative of a relationship rather than an exact measurement of it. The earliest two seasons in the data correspond to the *Premier League's* 'COVID' seasons; these were disrupted by the pandemic through a break in play and reduced fan presence. Season fixed-effects should absorb these league-wide effects.

Methodology:

To explore the relationship between league points and player wages, an OLS linear regression was used. League points function as a cardinal outcome with a wide range, therefore treating them as a continuous dependent variable makes linear regression methodologically appropriate. Season fixed effects ensure a club's residual (prediction error) reflects its performance measured against the season-specific expected points for its wage level, rather than a common baseline across all seasons.

A logarithmic transformation was applied to the annual wage bills in order to linearise the wage-points relationship and satisfy the linearity assumption of OLS. Theoretically, the return on wages is expected to display diminishing returns: a £50 million increase for a club with a £70 million wage bill should have a larger impact on its points tally than the same increase for a club already spending £200 million. Logging expresses the relationship in proportional terms, preventing the small number of extreme high-spenders from having an outsized influence on the model and improving the interpretability of the wage coefficient which now describes the points associated with a proportional change in wages. The residual-vs-fitted diagnostic (Figure 1) shows no systematic curvature, which suggests that the log-linear form chosen was appropriate. Cluster-robust standard errors, clustered by club, were used because clubs appear in multiple seasons which means observations are not independent. Ordinary standard errors would therefore violate OLS's independent-errors assumption, overstating the statistical significance of the results. This change only adjusts standard errors/p-values and leaves coefficients and R^2 unchanged. The choice to cluster standard errors rather than add club fixed-effects was conscious: clustering corrects uncertainty whilst keeping the between-club differences that are the central focus of this study. Club fixed-effects would absorb these differences, and in the persistence model that follows, their combination with a lagged dependent variable would introduce Nickell bias.

Diagnostic / Assumption Checks:

Four residual plots (residuals-vs-fitted, Q-Q, scale-location, residuals-vs-leverage) were used to test linearity, normality of residuals, homoscedasticity and influential points:

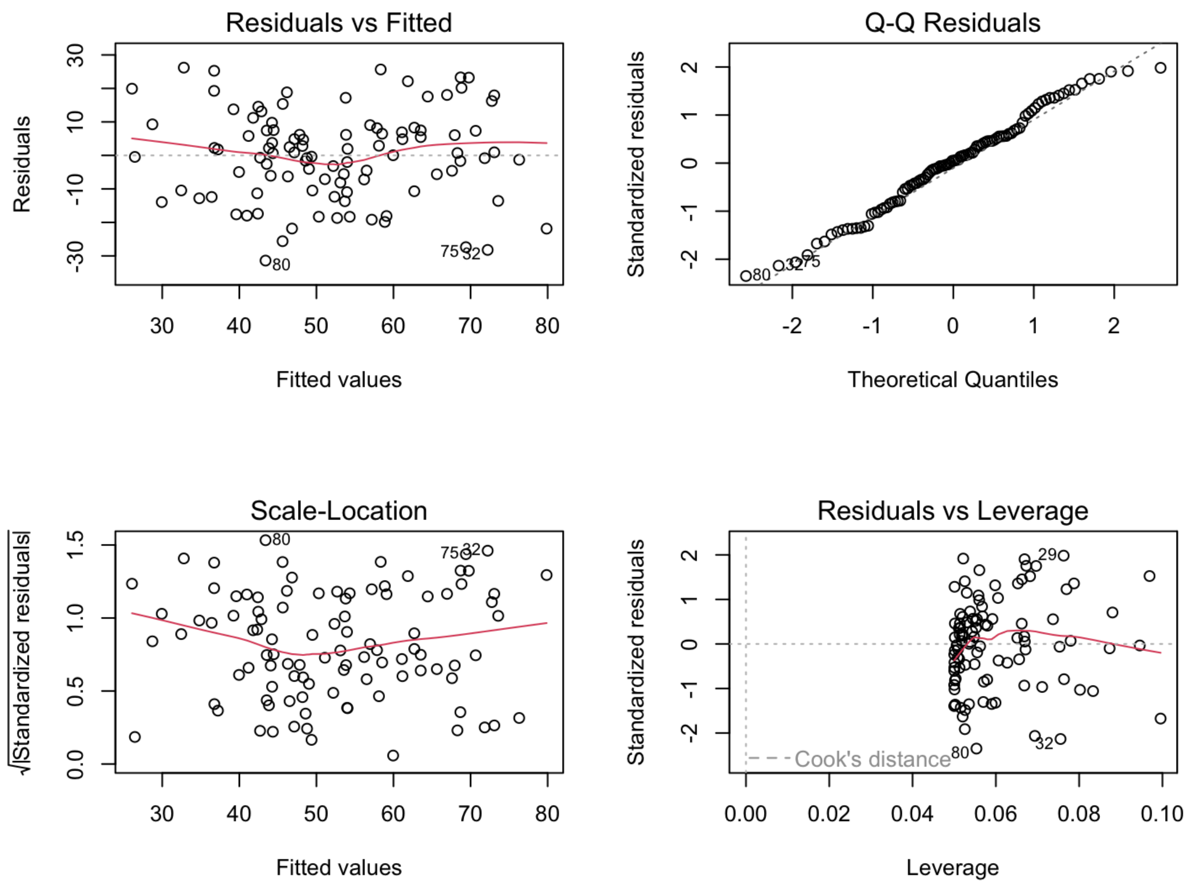

Figure 1: Regression diagnostic plots for the points-wage model
Clockwise from top-left: residuals-vs-fitted values, normal Q-Q, residuals-vs-leverage, and scale-location. Testing linearity, normality of residuals, influential observations, and homoscedasticity respectively.

No material violations observed: no systematic curvature, material departure from normality, or points beyond Cook's distance, hence supporting the OLS assumptions and the log-linear specification.

Residual:

The residuals from the points/wage regression are a central focus of this study. Defined as: $actual\ -\ predicted\ points$, a residual corresponds to the performance not explained by player wages.

Importantly, these residuals do not cleanly isolate efficiency or skill; they capture a mixture of durable non-wage factors, non-player spending and random noise. However, by testing whether the residuals persist from one season to the next, it is possible to infer the extent to which they are driven by noise/chance rather than genuine, non-player wage factors. If a club's residuals display year-on-year consistency, chance alone cannot account for them: indicating that real, non-player wage determinants contribute to its over- or under-performance relative to its wage bills.

Persistence Regression:

The logic underpinning the next model is simple: *luck does not repeat, skill does*. Residuals from the points/wage regression are modelled against the residuals from the same club's previous season. Clubs without consecutive seasons, typically those promoted or relegated, fall out of the model, which mildly skews the sample towards stable clubs.

A coefficient near 1 would indicate that the residuals entirely represent durable non-player wage factors, while a coefficient near 0 would suggest residuals purely represent noise.

As before, no club fixed-effects were used: the between-club differences are the focus of this study, and combining club fixed-effects with a lagged dependent variable would introduce Nickell bias in such a short panel (few seasons per club). Standard errors were again clustered by club.

Points/Wage Regression Results:

The OLS regression with league points tally as the dependent variable, annual player-wage spending as the explanatory variable and with season-fixed effects was fitted as follows (see Appendix A for full output):

$$League\ Points\ =\ -\ 340.9\ +\ 21.80 \cdot ln(Wage)\ +\ [season\ fixed\ effects]$$

$$Log\ Wage\ Coefficient: p\ =\ 3.42e-08\ \textit{(cluster-robust, clustered by club)}$$
$$Multiple\ R^2\ =\ 0.465$$
$$n\ =\ 140$$

The slope of the statistically significant ($p\ =\ 0.0000000342$) log-wage coefficient reads *21.8* so we can infer that a doubling of a *Premier League* club's wage spending is associated with a points increase of:

$$21.80\ *\ ln(2)\ =\ 15.111\ points$$

A 20% increase in a club's wage bill is associated with:

$$21.80\ *\ ln(1.2)\ =\ 3.975 \text{ extra points, } \textit{ceteris paribus}$$

The positive relationship between wage spending and points tally supports existing literature and the widely-held understanding of the game.

It is important to note that the R^2 value of *~0.47* means that most (~53%) of the observed variation in points tally is not explained by wage spending: this is an expected result for a single-season, single-league and single-variable study, and expected when looking holistically at the inherently low-scoring, highly volatile sport that football is.

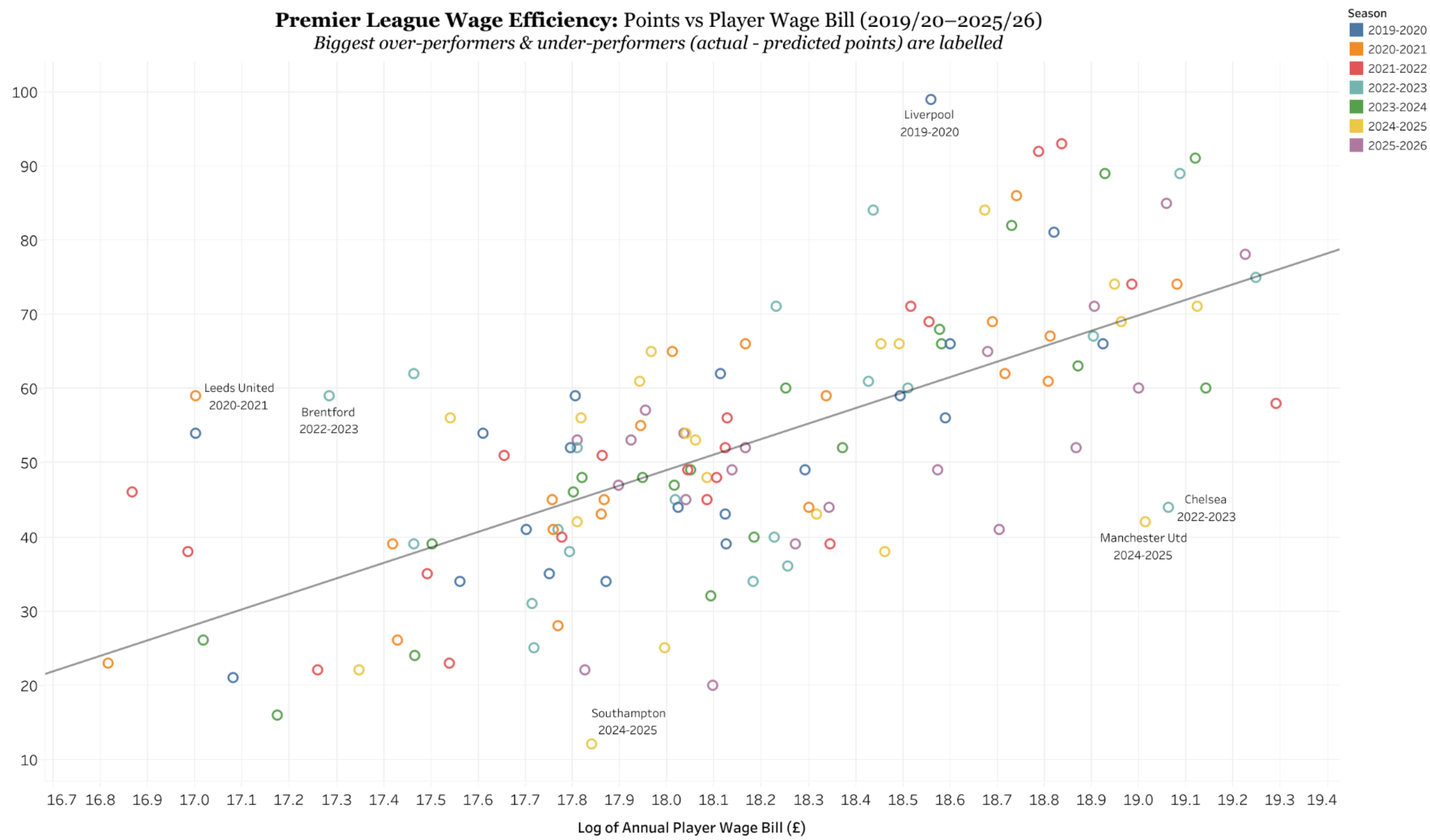

Figure 2: Premier League points against log annual player-wage bill (2019/20–2025/26) n = 140 club-seasons
The fitted line shows the season-adjusted wage–points relationship; a club's vertical distance from it is its wage efficiency (residual).
The three largest over- and under-performances are labelled.
Player wages estimated (FBref/Capology)

League points are plotted against log-wage and we can observe a positively-sloped relationship. The 3 biggest over- and under-performances are labelled and determined by the most positive and most negative residuals: the residual capturing a club's 'wage efficiency' (performance not explained by player wages). *Liverpool* (2019/20) beat their wage-derived expectations the furthest out of all 140 observed club-seasons, surpassing points expectations by 35.33 points (though this was the COVID-disrupted season) with *Brentford* (2022/23) and *Leeds* (2020/21) demonstrating that outperforming wage expectations is not exclusive to the biggest clubs.

Similar conclusions can be drawn in the opposite direction: with *Manchester United* (2024/25), *Chelsea* (2022/23), and *Southampton* (2024/25) most dramatically failing to meet their wage-derived predictions. Whether these reflect genuine operational failings or simply poor fortune is precisely what the persistence analysis will address.

The residuals of the regression ($actual\ points\ tally\ -\ wage\ predicted\ points\ tally$) need to be examined further before we can be confident in drawing any conclusions related to non-wage performance/'wage efficiency'. Noise exists in data, be that instances of chance, luck or misfortune: the residuals will unavoidably carry this variation that occurred due to mere chance, so one could just as easily conclude that clubs outperformed expectations simply due to luck as it was due to effective club operations.

Persistence Analysis Results:

Examination of persistence allows for determination regarding whether the residuals of the previous regression can be used for non-player wage performance interpretation. Luck and chance are random, so if over- and under-performance carries over season-to-season in a statistically significant manner, it points to something durable: be that structural or operational advantages (recruitment, coaching, data infrastructure, ownership, tactical cohesion, *et cetera*) rather than chance. See Appendix B for full output of the model.

Persistence was tested using a first-order autoregression (AR(1)):

$$E_{i,t} = \alpha + \rho \cdot E_{i,t-1} + u_{i,t}$$

where $E_{i,t}$ is club i's 'wage efficiency' (residual) in season t, ρ is the persistence coefficient, α is the intercept and $u_{i,t}$ is the error term: the portion of a season's 'efficiency' unrelated to the previous season's.

Fitted*:*

$$\hat{E}_{i,t} = -0.505 + 0.383 \cdot E_{i,t-1}$$

$$E_{i,t-1}\ Coefficient:\ p = 5.05e-05\ \textit{(cluster-robust, clustered by club)}$$

$$Multiple\ R^2 = 0.147$$

$$n = 112$$

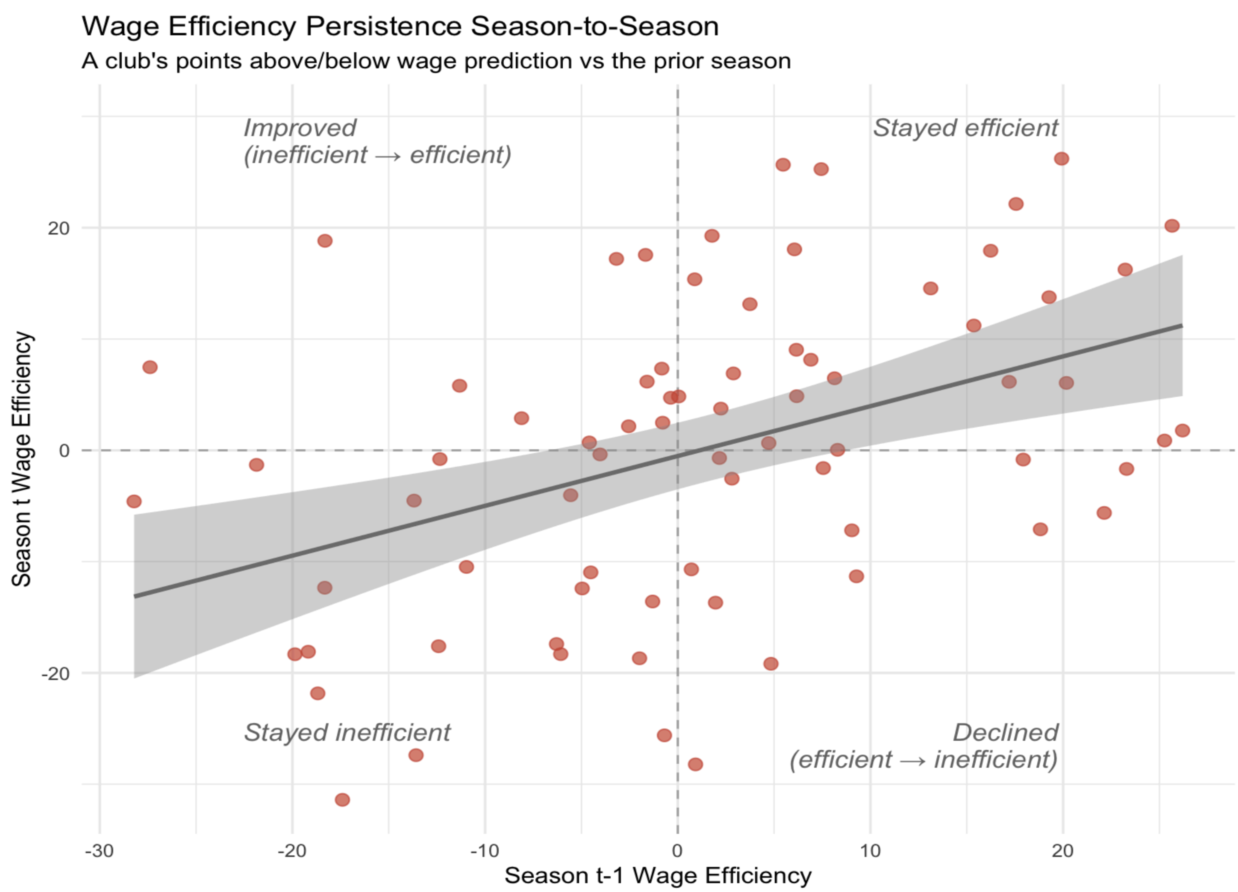

Figure 3: Season-to-season persistence of wage efficiency (n = 112 pairs)
Each point is a club-season's residual plotted against the same club's previous-season residual
Persistence coefficient = 0.38 (p < 0.001, cluster-robust), R^2 = 0.147

The results assign statistical significance ($p = 0.0000505$) to the residual's persistence, suggesting that the over- or under-performances are not purely due to chance. On average, roughly 38% of a club's over- or under-performance carries into the next season, so a club that beat its wage-derived expectations by +10 points last season is predicted to beat them by:

$$-\ 0.505\ +\ 0.383(10)\ =\ 3.33\ \textit{points the next season}$$

This is an average tendency so is not to be considered a firm rule: the low R^2 (0.147) reflects how previous-season over- or under-performance explains only ~15% of the variation in the following season's. 'Wage efficiency' carries over on average, but any single club's next season remains highly uncertain. The significant coefficient and low R^2 are not in tension as the former establishes that the relationship exists, the latter that it is loose.

Since the coefficient is significant and greater than 0, we can conclude that the over- and under-performances are not purely noise: a durable component exists. However, the coefficient is also well below 1, so these durable factors are only a partial explanation with chance and transient factors still accounting for the majority of any single season's deviation, but because over- and under-performance do persist, clubs that consistently outperform can be described as efficient rather than merely fortunate:

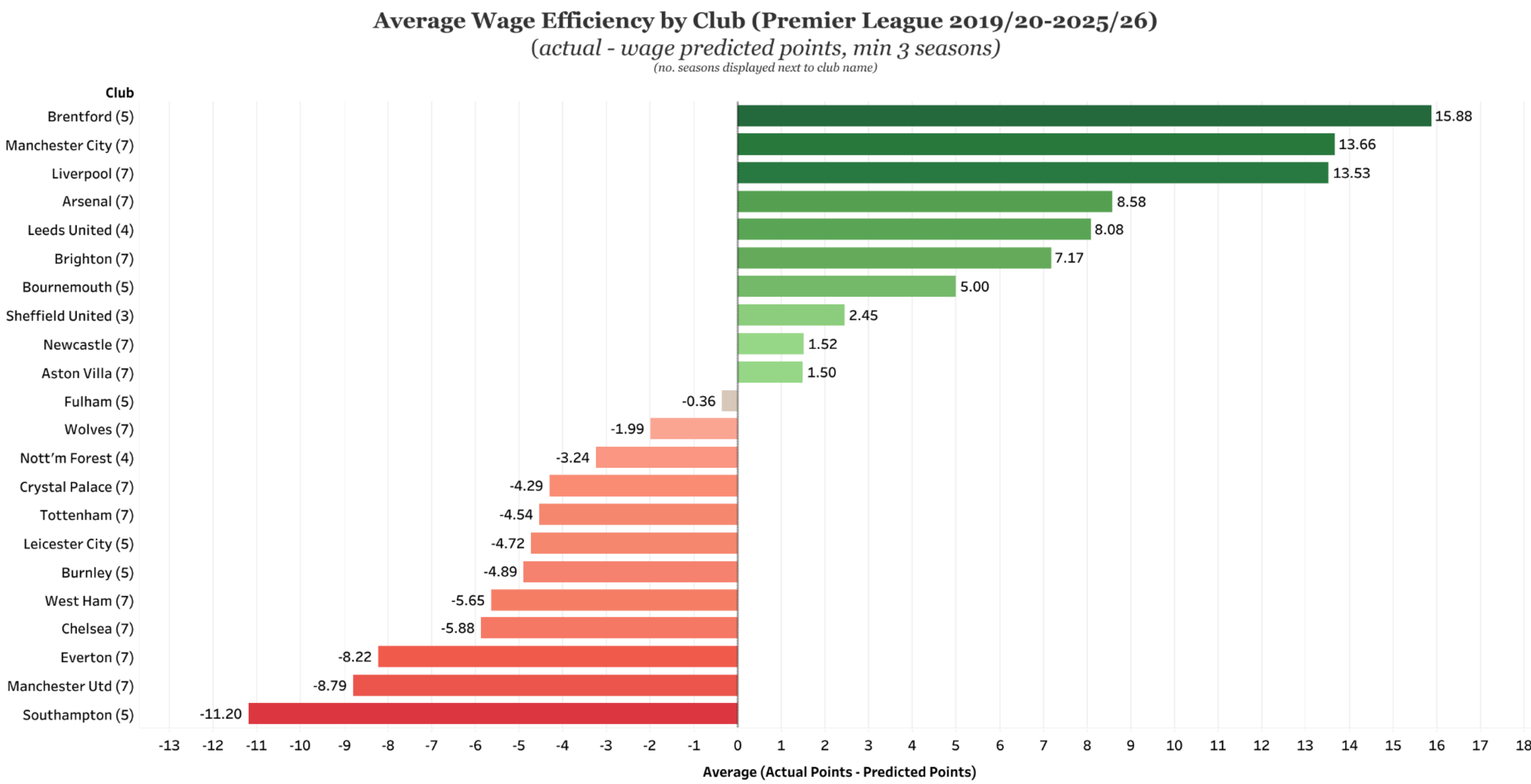

Figure 4: Mean wage efficiency by club, 2019/20–2025/26 (minimum three seasons)
Positive values indicate points earned above the wage-predicted total. As efficiency was shown to persist, this ranking reflects real advantages held during this period rather than chance alone.

Brentford appear to be the most operationally astute team in the *Premier League*, in this period, followed by *Manchester City* and *Liverpool*. On average, these 3 clubs beat their wage-derived points predictions by between 13-16 points per season. The driving factors behind this outperformance likely vary by club as *Manchester City* and *Liverpool* had two of the most prestigious managers in football: *Pep Guardiola* and *Jürgen Klopp*, which probably contributed to their over-performance; whilst *Brentford* famously adopt a bespoke, data-driven statistical approach to their footballing operations, which many prominent voices in football pin to their success.

Southampton displayed the worst non-wage performance, followed by *Manchester United* and *Everton*. This persistent underperformance suggests that these clubs' shortfalls are unlikely to be completely down to misfortune, and are consistent with durable, club-level attributes.

General Limitations:

Player Wages are not Total Spend: This research used player wages as the explanatory variable, so the residual absorbs all non-player spending: coaching salaries, backroom, academy, facilities, data infrastructure, *et cetera*. Because of this, a club could top the "wage-efficiency" ranking, simply by outspending rivals in non-player areas and reaping the benefits. A proportion of what this study calls "wage-efficiency" might therefore be spending that the model does not see, which is why a conscious effort was made to avoid conclusions regarding holistic spending, as the model concerns only one type of expenditure.

Measurement Error in the Wage Data: The wage figures carry an accuracy caveat: 'Capology' relies on estimates, and even its 'verified' figures are not drawn from actual contracts, with the estimated share varying by club and season. The key independent variable is therefore measured with error, which biases the wage coefficient toward zero and adds noise to every residual. Crucially, the direction of this error is likely to be club-specific rather than random. Although the share of a club's wage bill that is estimated varies year to year, the method underlying those estimates does not, so a club whose unverified figures are systematically overstated is likely to be overstated across seasons, even as the magnitude fluctuates. This matters for the persistence analysis: a consistent error in a club's wage figure produces a residual that is steadily displaced in the same direction for that club, which is observationally identical to a durable non-wage advantage or disadvantage. Some portion of the estimated persistence may therefore reflect durable *mismeasurement* rather than durable *quality*. Nothing in this design can separate the two, though the fact that the estimated share varies across seasons limits how much of the observed persistence this could plausibly account for; the finding that a durable component exists should nonetheless be read with that caveat attached.

Endogeneity / Reverse Causation: Wages and success are not purely unidirectional: just as higher wages support better performance, success feeds back into higher future wages through prize money, increased revenue, and larger commercial and image-rights deals. This introduces a degree of endogeneity, meaning that the wage coefficient doesn't capture a clean causal effect. The study is therefore descriptive rather than causal, and the coefficient should not be interpreted as the pure causal impact of spending on points.

COVID-disrupted Seasons: Two of the seven seasons *(2019/20, 2020/21)* were disrupted by the pandemic; empty stadiums removed home advantage and fixtures were congested. Season fixed-effects absorb the league-wide effect but any uneven club-level impact would leak into residuals.

Single-league Scope: As only the *Premier League* was examined, the findings of this study may not generalise to leagues with different competitive structures, wage distributions or financial regulations.

Modest Sample: The sample of 140 club-seasons and 112 pairs is modest; a longer panel would tighten the estimates.

Points/Wage Regression Limitations:

Unmodelled Performance Factors: Many factors affect a team's performance that are not captured by this model: be that coaching; staff recruitment, scouting and tactical coherence, just to name a few. While a high R^2 is often taken as a marker of a well-specified model, it was neither expected nor sought here. With only one explanatory variable (player-wage spending), and given that the object of the study is precisely the *unexplained* performance, a high R^2 was never the goal. The causes of these over- or under-performances are not in the scope of this study, and would likely vary by club. This study can isolate the non-player wage performance, not delve into its components.

Persistence Analysis Limitations:

Limited Predictive Power: The model shows that persistence exists but cannot predict a specific club's next season's efficiency with any accuracy ($R^2 = 0.147$). Though the model determines a relationship exists, it should not be used for forecasting.

Partial Durability: With a coefficient of ~*0.38*, the majority of any single season's deviation does not persist; the analysis confirms a durable component but efficiency is far from deterministic.

Survivorship in the Paired Sample: Clubs without consecutive seasons (those promoted/relegated) drop out of the paired sample, which comprises 112 pairs drawn from 24 of the 28 clubs, skewing it towards more stable clubs and limiting generalisability to "*yo-yo*" clubs that cycle between promotion and relegation.

Short Panel: With up to 7 seasons per club, the short panel constrains the precision of the estimate, and directed the avoidance of fixed effects (to prevent Nickell bias).

Unidentified Driver: While persistence shows that something durable persists, what that is specifically cannot be determined: it may reflect operational quality, but also durable non-player investment or other stable club characteristics that the model cannot separate, including persistent error in the wage estimates themselves (see *Measurement Error in the Wage Data*).

Conclusion:

This study set out to examine what the well-established link between wage spending and football success leaves unexplained. By focussing on the season-level variation that longer-run analyses average away, it found that player wages explain under half ($R^2 \approx 0.47$) of a single season's points variation, leaving unexplained performance that varies noticeably across clubs. Crucially, this component persists: a first-order autoregression showed that roughly *38%* of a club's over- or under-performance carries into the following season, remaining statistically significant after clustering standard errors by club. This rules out chance as the sole driver: durable, non-player wage determinants genuinely exist, but with the coefficient sitting well below *1* (~ 0.38), this persistence is only partial; with transient factors and noise still dominating any individual season.

Practically, this establishes that certain clubs, particularly *Brentford*, *Manchester City* and *Liverpool*, possess a repeatable ability to outperform their wage bills: consistent with durable operational quality; while others persistently underperform. The specific causes of these advantages cannot be isolated, but they are demonstrably real and measurable. In doing so, the study converts a widely-held belief that some clubs are simply "*better run*" into a quantified finding, and ultimately offers a public-data measure of footballing efficiency that could inform club valuation, between-club comparisons and recruitment analysis.

Further Work:

Replicating this two-stage study across Europe's top five leagues individually offers an extension of this research: enabling comparisons between the degree of 'wage-efficiency' persistence across Europe's differing competitive and financial structures. Isolating the top and worst performers across the continent can provide a springboard for deeper analysis that attempts to decompose the residuals; incorporating explanatory variables (managerial experience, squad age, net-transfer-spend, coaching stability, recruitment strategies, *et cetera*) to attribute the efficiency to specific drivers.

Player wages are not the same as total spend, so a natural extension would substitute total staff cost for estimated player wages. Since the current residual absorbs all non-player spending, part of what is labelled 'efficiency' may in fact be financial: a club outspending rivals on coaching, facilities or infrastructure. Placing total spend in the regressor would move more of the financial dimension out of the residual, leaving prediction errors that more closely reflect genuine non-financial drivers. This would better align the measure with the intended concept of efficiency.

Intuitively, extending the panel to include more seasons would enhance the analysis through increased consecutive-season pairs, more precise persistence estimates and testing whether persistence itself is stable over time. Though this is constrained by data quality, as *Premier League* wage figures prior to *2019/20* are fully estimated.

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Appendix A: Points/Wage Regression Full Model Output:

OLS with season fixed effects, n = 140 club-seasons. Cluster-robust standard errors (by club):

```
> summary(fit)

Call:
lm(formula = points ~ ln_wage + factor(season), data = model_df)

Residuals:
    Min      1Q  Median      3Q     Max 
-31.556  -8.478  -0.672   7.765  35.330 

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)             -340.90850   36.83821  -9.254 5.08e-16 ***
ln_wage                   21.79924    2.03502  10.712  < 2e-16 ***
factor(season)2020-2021   -0.03567    4.23236  -0.008   0.9933    
factor(season)2021-2022   -0.23143    4.23231  -0.055   0.9565    
factor(season)2022-2023   -2.76317    4.24146  -0.651   0.5159    
factor(season)2023-2024   -3.15719    4.24173  -0.744   0.4580    
factor(season)2024-2025   -4.46449    4.25214  -1.050   0.2957    
factor(season)2025-2026   -7.87823    4.28631  -1.838   0.0683 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.38 on 132 degrees of freedom
Multiple R-squared:  0.4652,	Adjusted R-squared:  0.4368 
F-statistic:  16.4 on 7 and 132 DF,  p-value: 2.013e-15
```

Cluster-robust standard errors, clustered by club:

```
> coeftest(fit, vcov = vcovCL, cluster = ~club)


t test of coefficients:


                          Estimate Std. Error t value  Pr(>|t|)    
(Intercept)             -340.90850   66.14262 -5.1541 9.080e-07 ***
ln_wage                   21.79924    3.71704  5.8647 3.421e-08 ***
factor(season)2020-2021   -0.03567    3.79185 -0.0094    0.9925    
factor(season)2021-2022   -0.23143    2.95083 -0.0784    0.9376    
factor(season)2022-2023   -2.76317    4.97992 -0.5549    0.5799    
factor(season)2023-2024   -3.15719    3.74825 -0.8423    0.4011    
factor(season)2024-2025   -4.46449    4.67693 -0.9546    0.3415    
factor(season)2025-2026   -7.87823    5.13499 -1.5342    0.1274    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Appendix B: Persistence Regression Full Model Output:

AR(1): efficiency ~ efficiency_prev, n = 112 pairs:

```
> summary(persist_fit)

Call:
lm(formula = efficiency ~ efficiency_prev, data = persistence_df)

Residuals:
     Min       1Q   Median       3Q      Max
-27.8672  -9.2725  -0.3705   7.8721  26.2535

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -0.50474    1.12618  -0.448    0.655
efficiency_prev  0.38322    0.08797   4.356 2.98e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.87 on 110 degrees of freedom
Multiple R-squared:  0.1471,     Adjusted R-squared:  0.1394
F-statistic: 18.98 on 1 and 110 DF,  p-value: 2.982e-05
```

Cluster-robust standard errors, clustered by club:

```
> coeftest(persist_fit, vcov = vcovCL, cluster = ~club)

t test of coefficients:

                 Estimate Std. Error t value  Pr(>|t|)
(Intercept)     -0.504739   1.295447 -0.3896    0.6976
efficiency_prev  0.383217   0.090814  4.2198 5.047e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```